

DEVIN WONG

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Education

University of California, San Diego

Bachelor of Science, Mechanical Engineering

Cumulative GPA: **3.70** / 4.00

Available for an internship from **June 2027** to **September 2027**

Expected June 2028

La Jolla, CA

Experience

Tesla

Sep 2026 – Jun 2027

Tooling & Process Engineering Intern - Castings

Lathrop, CA

- Identified cost-saving process changes to reduce cycle times and improve product quality
- Worked closely with technicians on preventive maintenance and day-to-day production operations
- Collaborated with Tooling, Process, Production, and Quality teams to deploy changes
- Supported RCA and CAPA actions to resolve casting-quality issues and prevent recurrence
- Applied GD&T and tolerance stack-ups to improve casting dimensional capability

Roche

Jun 2026 – Sep 2026

Mechanical Engineering Intern - Sensor Module Development

Santa Clara, CA

- Owned a trade study of **four** sensor-module geometries, selecting the one that cut shear **70%** and pressure drop **72%**
- Built grid-independent CFD in **SolidWorks Flow Simulation (15M cells)**, validated against a **PD-pump** rig
- Ran steady-state thermal analysis across sensor interfaces, improving temperature uniformity by a **factor of two**
- Designed and fabricated pipette-tip fixtures and test stands to accelerate fluidic-interface prototyping

Triton Racing (Formula SAE) - University of California, San Diego

Oct 2025 – Jun 2026

Mechanical Design & Analysis Engineer - Aerodynamics

San Diego, CA

- Owned the TR26 rear-wing structures package end-to-end, setting load cases and driving material/layout trade studies
- Performed structural FEA in **ANSYS Mechanical**, validating parts under **200+ lb** loads to a target **FOS ~1.5**
- Conducted 2D steady-state CFD in **ANSYS Fluent** to optimize pressure distributions, slot gaps, and element sizing
- Generated CAM toolpaths in **Fusion 360** to machine high-density foam molds on a 3-axis CNC router
- Authored **10+** detailed **GD&T** drawings (ASME Y14.5-2018), improving fabrication and alignment accuracy

Projects

Triton Racing | Front Wing 2D CFD | ANSYS Fluent, Verification & Validation, Parametric Optimization **Mar 2026**

- Ran a two-element airfoil study (**NACA 4412 + S1223**) in **ANSYS Fluent** (k- ω SST) to **Cl 2.74**, **L/D 11.0**
- Established mesh independence via a four-level study and Richardson-extrapolation **GCI of 2.4%**
- Drove design with a **50-point** sweep across **5** parameters, identifying element spacing as the dominant sensitivity

Triton Racing | Rear Wing Spar | SolidWorks, ANSYS ACP, Fusion 360, CNC Router, Wet Layup **Jan 2026**

- Designed a **31"** carbon-fiber spar carrying a **400 N/m** distributed aerodynamic design load, validated to a FOS of **5.5**
- Created CAM toolpaths to CNC-machine a foam mold to **0.004"** stepdown tolerance, enabling clean release post layup
- Conducted a **4-ply** bidirectional carbon-fiber layup holding peak deflection to **0.38 mm** under aerodynamic load
- Validated laminate stiffness against ANSYS Composite Analysis with a physical **3-point bend test**

Triton Racing | Rear Wing Mounts | SolidWorks, ANSYS Mechanical, FEA, Bolt Theory, Water Jet **Nov 2025**

- Cut mount weight by **77%** via topology optimization of **6061-T6** swan-neck mounts while meeting the load case
- Verified FEA against hand calculations (bending stress, deflection at **200 N**) for a first-principles baseline
- Ran a **5-level, 2.2M-cell** mesh study to **<0.5%** variation, resolving BC singularities to read true peak stress
- Sized the chassis-interface bolted joint (shear/bearing) and fabricated final mounts on a water jet

Technical Skills

Software: SolidWorks, SolidWorks Flow Simulation, CATIA V5, 3DEXPERIENCE, ANSYS Mechanical, ANSYS ACP, ANSYS Fluent, Fusion 360

Design & Analysis: GD&T (ASME Y14.5-2018), DFM/DFA, CAD, CAM, FEA, CFD, Mesh Convergence, MATLAB

Problem Solving Frameworks: RCA, CAPA, 8D, Fault Tree Analysis (FTA), Fishbone (Ishikawa)

Machining: CNC Router, Laser Cutter, 3D Printer, Water Jet, Hot-Wire Cutter